Ibn Tofail University

Analysis V

Problem Set II

Exercise 1:

Study the existence of a limit at $(0,0)$ for the following functions:

$f(x,y) = \dfrac{x}{y}$; $f(x,y) = \dfrac{1-\cos(xy)}{y^2}$; $f(x,y) = \dfrac{(x^2+y^2)^2}{x^2-y^2}$

$f(x,y) = \dfrac{2x^2+xy}{\sqrt{x^2+y^2}}$; $f(x,y) = \dfrac{\sin(xy)}{|x|+|y|}$; $f(x,y) = \begin{cases} \dfrac{|xy|^\alpha}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$, where $\alpha > 0$

Correction

1. $f(x,y) = \dfrac{x}{y}$.

   Conclusion: The limit does not exist.

   Proof: Along the line $y = x$ (with $x \to 0$) we get $f(x,x) = 1$. Along the line $x = 0$ (with $y \to 0$) we get $f(0,y) = 0$. Different path-limits, so the two-variable limit does not exist.

2. $f(x,y) = \dfrac{1-\cos(xy)}{y^2}$.

   Conclusion: The limit exists and equals 0.

   Proof: Using $1 - \cos t = 2\sin^2(t/2)$ and $|\sin u| \leq |u|$,

   $$\left|1-\cos(xy)\right| = 2\sin^2\frac{xy}{2} \leq 2\Big(\frac{|xy|}{2}\Big)^2 = \frac{(xy)^2}{2}.$$

   Hence, for $y \neq 0$,

   $$\left|\frac{1-\cos(xy)}{y^2}\right| \leq \frac{x^2y^2}{2y^2} = \frac{x^2}{2}.$$

   Since $x^2/2 \to 0$ as $(x,y) \to (0,0)$, we conclude $\displaystyle\lim_{(x,y)\to(0,0)} \frac{1-\cos(xy)}{y^2} = 0$.

3. $f(x,y) = \dfrac{(x^2+y^2)^2}{x^2-y^2}$.

   Conclusion: The limit does not exist.

Proof: For any fixed slope $y = mx$ with $m \neq \pm 1$,

$$f(x, mx) = \frac{(x^2 + m^2x^2)^2}{x^2 - m^2x^2} = \frac{x^4(1 + m^2)^2}{x^2(1 - m^2)} = x^2 \cdot \frac{(1 + m^2)^2}{1 - m^2} \xrightarrow[x \to 0]{} 0.$$

Thus along every fixed non-characteristic straight line we get 0.

But we can find a Cartesian path that yields a different limit. Define a path by

$$y(x) = x\sqrt{1 - 4x^2} \qquad \text{for } |x| \leq \tfrac{1}{2}.$$

Then

$$x^2 + y(x)^2 = x^2\big(1 + (1 - 4x^2)\big) = x^2(2 - 4x^2),$$

so

$$(x^2 + y(x)^2)^2 = x^4(2 - 4x^2)^2, \qquad x^2 - y(x)^2 = x^2\big(1 - (1 - 4x^2)\big) = 4x^4.$$

Therefore for $x \neq 0$ with $|x| \leq \frac{1}{2}$,

$$f\big(x, y(x)\big) = \frac{x^4(2 - 4x^2)^2}{4x^4} = (1 - 2x^2)^2.$$

As $x \to 0$ this tends to 1. So along the path $y = y(x)$ the limit is 1, while along lines $y = mx$ (with fixed $m \neq \pm 1$) it is 0. Different limits $\Rightarrow$ no limit at $(0, 0)$.

4. $f(x, y) = \dfrac{2x^2 + xy}{\sqrt{x^2 + y^2}}$.

Conclusion: The limit exists and equals 0.

Proof: Let $R = \sqrt{x^2 + y^2}$. Then $|x| \leq R$ and $|y| \leq R$. Also $|xy| \leq \frac{x^2+y^2}{2} = \frac{R^2}{2}$. Hence

$$|2x^2 + xy| \leq 2x^2 + |xy| \leq 2R^2 + \frac{R^2}{2} = \frac{5}{2}R^2.$$

Therefore

$$\left|\frac{2x^2 + xy}{\sqrt{x^2 + y^2}}\right| \leq \frac{\frac{5}{2}R^2}{R} = \frac{5}{2}R,$$

which tends to 0 as $(x, y) \to (0, 0)$. So the limit is 0.

5. $f(x, y) = \dfrac{\sin(xy)}{|x| + |y|}$.

Conclusion: The limit exists and equals 0.

Proof: Use $|\sin t| \leq |t|$:

$$\left|\frac{\sin(xy)}{|x| + |y|}\right| \leq \frac{|xy|}{|x| + |y|}.$$

From $(|x|-|y|)^2 \geq 0$ we get $x^2+y^2 \geq 2|x||y|$, hence

$$|xy| \leq \frac{x^2+y^2}{2} \leq \frac{(|x|+|y|)^2}{4}.$$

Thus

$$\frac{|xy|}{|x|+|y|} \leq \frac{(|x|+|y|)^2}{4(|x|+|y|)} = \frac{|x|+|y|}{4} \xrightarrow[(x,y)\to(0,0)]{} 0.$$

Hence the given quotient tends to 0.

6.

$$f(x,y) = \begin{cases} \dfrac{|xy|^\alpha}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases} \qquad (\alpha > 0).$$

Conclusion:

$$\lim_{(x,y)\to(0,0)} f(x,y) = \begin{cases} 0 & \text{if } \alpha > 1, \\ \text{does not exist} & \text{if } \alpha = 1, \\ \text{no finite limit (unbounded)} & \text{if } 0 < \alpha < 1. \end{cases}$$

Proof: Consider the straight path $y = mx$ (with fixed $m \in \mathbb{R}$). For $x \neq 0$,

$$f(x,mx) = \frac{|x \cdot mx|^\alpha}{x^2+m^2x^2} = \frac{|m|^\alpha |x|^{2\alpha}}{x^2(1+m^2)} = \frac{|m|^\alpha}{1+m^2}\,|x|^{2\alpha-2}.$$

- If $\alpha > 1$, then $2\alpha - 2 > 0$ so $|x|^{2\alpha-2} \to 0$ as $x \to 0$. The factor $\dfrac{|m|^\alpha}{1+m^2}$ is bounded in m, hence $f(x,mx) \to 0$ for every m. Moreover the bound is uniform in direction because

$$\frac{|xy|^\alpha}{x^2+y^2} \leq \frac{(x^2+y^2)^\alpha}{x^2+y^2} = (x^2+y^2)^{\alpha-1},$$

  and when $\alpha > 1$ the right-hand side tends to 0. So the limit is 0.
- If $\alpha = 1$, then

$$f(x,mx) = \frac{|m|}{1+m^2},$$

  which depends on m. For $m = 0$ the value is 0, for $m = 1$ the value is $1/2$. Different directional limits $\Rightarrow$ no limit.
- If $0 < \alpha < 1$, then $2\alpha - 2 < 0$. For any fixed m with $m \neq 0$,

$$f(x,mx) = \frac{|m|^\alpha}{1+m^2}\,|x|^{2\alpha-2} \xrightarrow[x\to 0]{} +\infty,$$

  so the function is unbounded near $(0,0)$ and no finite limit exists.

Exercise 2:

Study the limit at $(0,0)$ of the two-variable numerical function f defined by:

$$f(x,y)=\frac{\cos x-\cos y}{x-y}$$

Correction

Method 1 — trigonometric identity and estimation (purely Cartesian). For $x\neq y$ use the identity

$$\cos x-\cos y=-2\sin\frac{x+y}{2}\,\sin\frac{x-y}{2}.$$

Hence

$$\frac{\cos x-\cos y}{x-y}=-2\sin\frac{x+y}{2}\cdot\frac{\sin\frac{x-y}{2}}{x-y}.$$

Write

$$\frac{\sin\frac{x-y}{2}}{x-y}=\frac{1}{2}\cdot\frac{\sin\frac{x-y}{2}}{\frac{x-y}{2}}.$$

Therefore

$$\frac{\cos x-\cos y}{x-y}=-\sin\frac{x+y}{2}\cdot\frac{\sin\frac{x-y}{2}}{\frac{x-y}{2}}.$$

Since for all real u we have $\left|\sin u/u\right|\leq 1$, it follows

$$\left|\frac{\cos x-\cos y}{x-y}\right|\leq\left|\sin\frac{x+y}{2}\right|.$$

As $(x,y)\to(0,0)$ we have $\sin\frac{x+y}{2}\to 0$. By the squeeze principle the quotient tends to 0. Thus

$$\boxed{\lim_{(x,y)\to(0,0)}\frac{\cos x-\cos y}{x-y}=0.}$$

Remark (optional, Cartesian MVT view). Fix a sequence $(x_n,y_n)\to(0,0)$ with $x_n\neq y_n$. By the one-variable Mean Value Theorem there exists c_n between x_n and y_n such that

$$\frac{\cos x_n-\cos y_n}{x_n-y_n}=-\sin c_n.$$

Since $c_n\to 0$ as $n\to\infty$, $-\sin c_n\to 0$. This gives the same limit 0.

Exercise 3:

Study the existence of a limit at the indicated point for the following functions:

$$f(x,y,z)=\frac{xyz}{x+y+z}\text{ at }(0,0,0);\quad f(x,y,z)=\frac{x+y}{x^2-y^2+z^2}\text{ at }(2,-2,0)$$

Correction

1) $f(x,y,z) = \dfrac{xyz}{x+y+z}$ **at** $(0,0,0)$**.**

Claim. The limit $\displaystyle\lim_{(x,y,z)\to(0,0,0)} \frac{xyz}{x+y+z}$ does *not* exist.

Proof (explicit Cartesian path giving a nonzero limit). Let $a_n > 0$ be any sequence with $a_n \to 0$ (for instance $a_n = \frac{1}{n}$). Define

$$x_n = a_n, \qquad y_n = -\tfrac{1}{2}a_n, \qquad z_n = -\tfrac{1}{2}a_n + a_n^3.$$

Then $(x_n, y_n, z_n) \to (0,0,0)$ and

$$x_n + y_n + z_n = a_n\Big(1 - \tfrac{1}{2} - \tfrac{1}{2} + a_n^2\Big) = a_n^3,$$

while

$$x_n y_n z_n = a_n \cdot \Big(-\tfrac{1}{2}a_n\Big) \cdot \Big(-\tfrac{1}{2}a_n + a_n^3\Big) = a_n^3\Big(\tfrac{1}{4} + O(a_n^2)\Big).$$

Therefore

$$f(x_n, y_n, z_n) = \frac{x_n y_n z_n}{x_n + y_n + z_n} = \frac{a_n^3\big(\frac{1}{4} + O(a_n^2)\big)}{a_n^3} \longrightarrow \tfrac{1}{4}.$$

Thus along this Cartesian path the function values tend to 1/4.

Conclusion. Because one can construct sequences approaching $(0,0,0)$ that produce different limit values (for example many other paths give 0 while the path above gives 1/4), the limit does not exist.

2) $f(x,y,z) = \dfrac{x+y}{x^2 - y^2 + z^2}$ **at** $(2,-2,0)$**.**

Introduce small increments

$$u = x - 2, \qquad v = y + 2, \qquad w = z,$$

so $(u,v,w) \to (0,0,0)$ corresponds to $(x,y,z) \to (2,-2,0)$. Then

$$x + y = u + v,$$

and a direct algebraic expansion gives

$$x^2 - y^2 + z^2 = (2+u)^2 - (-2+v)^2 + w^2 = 4(u+v) + \big(u^2 - v^2 + w^2\big).$$

Therefore for $(u,v,w) \neq (0,0,0)$ with $u + v \neq 0$,

$$f(x,y,z) = \frac{u+v}{4(u+v) + (u^2 - v^2 + w^2)} = \frac{1}{4 + \dfrac{u^2 - v^2 + w^2}{u+v}}.$$

Two Cartesian approaches giving different limits.

- **Line with $v = mu$ and $w = 0$ (any fixed $m \neq -1$).** Set $v = mu$, $w = 0$. Then $u + v = (1+m)u$ and

$$u^2 - v^2 + w^2 = u^2 - m^2u^2 = (1 - m^2)u^2.$$

Thus

$$f(x,y,z) = \frac{(1+m)u}{4(1+m)u + (1-m^2)u^2} = \frac{1}{4 + u \cdot \frac{1-m^2}{1+m}} \longrightarrow \frac{1}{4} \quad \text{as } u \to 0.$$

So along any such straight line (except the degenerate $m = -1$) the limit is $1/4$.

- **Path with $v = -u,\ w = u$.** Take $v = -u$ and $w = u$. Then $u + v = 0$ and

$$u^2 - v^2 + w^2 = u^2 - u^2 + u^2 = u^2,$$

so

$$f(x,y,z) = \frac{u+v}{4(u+v) + (u^2 - v^2 + w^2)} = \frac{0}{u^2} = 0$$

for $u \neq 0$. Hence along this Cartesian path the function value is 0, and the limit along this path is 0.

Conclusion. Two Cartesian approaches give different limits (1/4 and 0), so the full three-variable limit at $(2, -2, 0)$ does not exist.

Exercise 4:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$f(x,y) = \begin{cases} 2y & \text{if } x \geq y^2 \\ \dfrac{2x}{y} & \text{if } |x| < y^2 \text{ and } y \neq 0 \\ -2y & \text{if } x \leq -y^2 \end{cases}$$

Study the continuity of f on $\mathbb{R}^2$.

Correction

Claim

f is continuous at every point of $\mathbb{R}^2$.

Proof

Clearly $2y$, $2x/y$ (on its domain $y \neq 0$) and $-2y$ are continuous on their respective open regions. We must check continuity on the boundaries $x = y^2$, $x = -y^2$, and on the x-axis $y = 0$.

1. Points on $x = y^2$ with $y \neq 0$. Let $(x,y) \to (y^2, y)$ with $y \neq 0$. From the right (region $x \geq y^2$) the value is $2y$. From the central region $|x| < y^2$ we have

$$\frac{2x}{y} \xrightarrow[x \to y^2]{} \frac{2y^2}{y} = 2y.$$

Hence both sides give the same limit $2y$, so f is continuous at (y^2, y).

2. Points on $x = -y^2$ with $y \neq 0$. Analogously, approaching $(-y^2, y)$ from the left (region $x \leq -y^2$) yields $-2y$, and from the central region

$$\frac{2x}{y} \xrightarrow[x \to -y^2]{} \frac{2(-y^2)}{y} = -2y.$$

Thus f is continuous at $(-y^2, y)$.

3. Points on the x-axis with $x_0 \neq 0$. Fix $x_0 \neq 0$. Choose $r > 0$ so small that $r < |x_0|/2$. If (x, y) satisfies $|(x, y) - (x_0, 0)| < r$ then

$$|x| \geq |x_0| - |x - x_0| > |x_0| - r \geq |x_0|/2.$$

Also $|y| < r$ so $y^2 < r^2 < (|x_0|/2)^2$. Hence for such points $|x| > y^2$, so the central region $|x| < y^2$ is excluded in this neighborhood. Therefore on a small neighborhood of $(x_0, 0)$ the formula for f is either $2y$ or $-2y$; in either case

$$|f(x, y) - f(x_0, 0)| = | \pm 2y - 0| \leq 2|y| \xrightarrow[(x,y) \to (x_0,0)]{} 0.$$

Thus f is continuous at every $(x_0, 0)$ with $x_0 \neq 0$.

4. The origin $(0, 0)$. We have $f(0, 0) = 0$. If (x, y) lies in the outer regions (where formula is $\pm 2y$) then $|f(x, y)| \leq 2|y| \to 0$. If (x, y) lies in the central region $|x| < y^2$ (and $y \neq 0$), then

$$\left|\frac{2x}{y}\right| \leq \frac{2|x|}{|y|} < \frac{2y^2}{|y|} = 2|y| \to 0.$$

Hence in every case $f(x, y) \to 0 = f(0, 0)$ as $(x, y) \to (0, 0)$. So f is continuous at the origin.

Combining the above, f is continuous at every point of $\mathbb{R}^2$.

Exercise 5:

Study the continuity on $\mathbb{R}^2$ of the functions defined in Exercises 2 and 3.

Correction

Same as the correction of Exercise 4.

Exercise 6:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$f(x, y) = \begin{cases} \dfrac{|x|^\alpha \cdot |y|^{2\alpha}}{|x|^\alpha + |y|^{2\alpha}} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Determine the values of α for which f is continuous at $(0, 0)$.

Correction

Claim

f is continuous at $(0,0)$ exactly for $\alpha > 0$.
Proof

Case $\alpha > 0$. Put

$$a := |x|^\alpha, \qquad b := |y|^{2\alpha} \qquad (a, b \geq 0).$$

For $a, b > 0$ we have

$$0 \leq \frac{ab}{a+b} \leq \min\{a, b\},$$

because if $a \leq b$ then $\dfrac{ab}{a+b} \leq \dfrac{ab}{b} = a$ (and symmetrically if $b \leq a$). Thus for all $(x, y) \neq (0, 0)$,

$$0 \leq f(x, y) \leq \min\{|x|^\alpha,\ |y|^{2\alpha}\}.$$

When $(x, y) \to (0, 0)$ both $|x|^\alpha \to 0$ and $|y|^{2\alpha} \to 0$ (since $\alpha > 0$), hence $\min\{|x|^\alpha, |y|^{2\alpha}\} \to 0$. By the squeeze theorem $f(x, y) \to 0 = f(0, 0)$. Therefore f is continuous at $(0, 0)$.
Case $\alpha = 0$. For $(x, y) \neq (0, 0)$ one has $|x|^\alpha = |y|^{2\alpha} = 1$, so

$$f(x, y) = \frac{1 \cdot 1}{1 + 1} = \tfrac{1}{2} \qquad \text{for } (x, y) \neq (0, 0),$$

while $f(0, 0) = 0$. Thus $\lim_{(x,y)\to(0,0)} f(x, y) = \frac{1}{2} \neq 0$, so f is discontinuous at $(0, 0)$.
Case $\alpha < 0$. Take the diagonal path $x = y \to 0$ (with $x \neq 0$). Then

$$|x|^\alpha = |y|^\alpha \to +\infty,$$

so $a = b \to +\infty$ and

$$f(x, x) = \frac{a^2}{2a} = \frac{a}{2} \xrightarrow[x \to 0]{} +\infty.$$

Hence f is unbounded near $(0, 0)$ and in particular there is no finite limit; f is not continuous at $(0, 0)$.